INCH-POUND
MIL-I-24718/4
6 April 1990

MILITARY SPECIFICATION SHEET

# INSULATING RESINS, SOLVENTLESS, VACUUM-PRESSURE-IMPREGNATING REPLENISHMENT RESIN, POLYESTER DIALLYL PHTHALATE, SLIGHTLY THIXOTROPIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-I-24718.

Resin: The product shall consist of primarily polyester resin with diallyl phthalate monomer, catalyst, and appropriate filler and thixotropes. The product shall be compatible with resin samples provided by the acquisition activity and meet all the requirements specified herein. The resin samples provided shall be of the same generic class (for example, polyester diallyl phthalate).

Thermal class: The cured resin mixtures (see compatibility section below) shall have thermal endurance that is not less than the thermal endurance of the sampled resin when tested in accordance with MIL-I-24718.

Rheology: The rheological properties of the resin mixtures shall be as specified and suitable for VPI of form and random wound electrical coils and windings.

Compatibility: The mixing tolerance of the catalyzed product shall be determined by using new resin prepared in accordance with MIL-I-24718 and resin supplied by the acquisition activity. The new resin shall be added to the supplied resin at 23°C, stirring slowly until mixtures of 9:1, 3:1, 1:1, 1:3, and 1:9 by weight are obtained. The mixtures shall then be examined for curdling, precipitation, or separation and shall be examined again after 24 hours. The resulting mixtures shall be homogeneous and shall not separate or contain gelatinous material or precipitate. Curing conditions shall be mutually agreed upon by the acquisition activity and the contracting agent. Each mix shall meet the first article requirements as specified herein. The replenishment product shall also be tested without addition of any other material.

AMSC N/A FSC 5970

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Requirements: The following property requirements shall be met for 100 percent replenishment product and all mixtures of replenishment product with supplied resin.

First article inspection. First article inspection shall be as specified in MIL-I-24718 and table I herein:

TABLE I. First article inspection.

| Characteristics | Unit | Requirements |
|---|---|---|
| Catalyzed storage life | Months at 20°C, minimum | 9 |
| Vapor pressure | mm Hg, maximum | 2.0 |
| Weight loss on cure | Average percent, maximum | 30.0 |
| Flash point | °C, minimum | 93 |
| Viscosity | | |
| at 2 r/min | Average centipoise, maximum | 15,000 |
| at 20 r/min | Average centipoise, maximum | 5,000 |
| Shear thinning index | Minimum | 1.10 |
| Build | Mils, minimum | 2.0 |
| | Mils, maximum | 5.0 |
| Resin retention | Weight percent, minimum | 75.0 |
| Dielectric strength | | |
| at 96/23/50 | Volts per mil, minimum average per panel | 1,000 |
| at 96/23/96 | Percent decrease from 96/23/50 value, maximum | 25 |
| at 24/23/water | Percent decrease from 96/23/50 value, maximum | 25 |
| Dissipation factor | | |
| at 23°C | Percent, maximum | 3.0 |
| at 150° | Percent, maximum | NA |
| Dielectric constant | | |
| at 23°C | Maximum | 5.0 |
| at 150°C | Maximum | NA |
| Bond strength | | |
| at 23°C | Pounds force, minimum | 30.0 |
| at 150°C | Pounds force, minimum | 5.0 |
| Thermal endurance, all classes | Thermogravimetric analysis comparison with sampled resin, °C temperature difference at any weight loss | minus 5 plus 0 |
| Hydrolytic stability | Percent decrease in bond strength, maximum | 50 |
| Hardness | Type D, minimum | 70 |
| Chemical resistance | | |
| Hydraulic fluid | Percent decrease in bond strength, maximum | 25 |
| Lubricating oil | Percent decrease in bond strength, maximum | 25 |



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TABLE I. First article inspection - Continued.

| Characteristics | Unit | Requirements |
|---|---|---|
| Cleaning fluid | Percent decrease in bond strength, maximum | 25 |
| Distilled water | Percent decrease in bond strength, maximum | 25 |

Quality conformance inspection. Quality conformance inspection shall be as specified in MIL-I-24718 and table II herein.

TABLE II. Quality conformance requirements.

| Characteristics | Unit | Requirements |
|---|---|---|
| Detergent solution | Percent decrease in bond strength, maximum | 25 |
| Weight loss on cure | Average percent, maximum | 30 |
| Gel point | Percentage of qualification value, plus or minus | 10 |
| Viscosity at 2 r/min | Average centipoise, maximum | 15,000 |
| | Percentage of qualification value, plus or minus | 10 |
| Build | Mils, minimum | 2.0 |
| | Mils, maximum | 5.0 |
| Resin retention | Weight percent, minimum | 75.0 |
| Specific gravity | Percentage of qualification value, plus or minus | 10.0 |
| Dielectric strength at 24/23/50 | Volts per mil, minimum | 1,000 |
| Hardness | Percentage of qualification value, plus or minus | 10.0 |
| Bond strength at 25°C | Pounds force, minimum | 30.0 |

Intended use: The product shall be used as replenishment resin for MIL-I-24718/3 type VPI resins only after direct approval from NAVSEA. It shall be tested alone and with sample storage resin supplied by the acquisition activity.



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Part or Identifying Number (PIN): The PIN shall be M24718/4-(dash number indicating thermal class). For example, M24718/4-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance.

Custodians:
Army - ER
Navy - SH
Air Force - 20

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
Army - MF
Navy - MC, AS

Preparing activity:
Navy - SH
(Project 5970-1002-04)



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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-I-24718/4 | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|

**DOCUMENT TITLE**

INSULATING RESINS, SOLVENTLESS, VACUUM-PRESSURE-IMPREGNATING: GENERAL SPECIFICATION FOR

**NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**REASON FOR RECOMMENDATION**

**SUBMITTER**

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
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| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**PREPARING ACTIVITY**

| a. NAME Technical Point of Contact (TPOC): Mr. C.Y. Lu (SEA 56Z24)<br>USE ADDRESS ALL CORRESPONDENCE AS FOLLOWS: | b. TELEPHONE (Include Area Code)<br>(1) Commercial: TPOC: 703-602-3123<br>(2) AUTOVON: 8-332-3123 |
|---|---|
| c. ADDRESS (Include Zip Code)<br>Commander, Naval Sea Systems Command<br>Department of the Navy<br>Washington, DC 20362-5101 | 8. IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

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